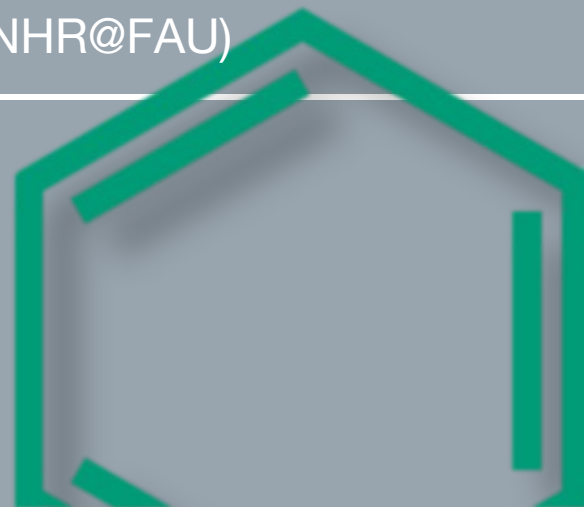


Diagnosing Overhead in Dispatch Operations: Cross-architecture Observatory

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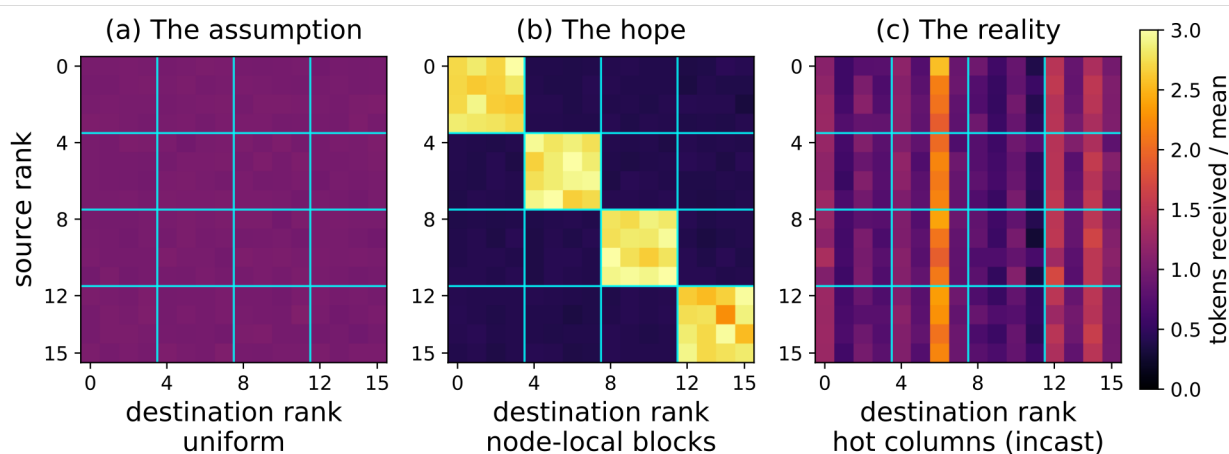
What actually decides MoE latency?

- Mixture of Experts stresses differently from classic HPC workloads
 - In an LLM, the experts (FFN) hold most of the weights
 - Attention weights, the other major chunk, are comparatively small
- HBM per GPU is limited: the experts cannot all sit on one device
 - Sharding the experts is what brings in All-to-All
 - Scale-up or more often scale-out
- Every MoE block into two AlltoAll collectives
 - A training step spends roughly half its forward-pass time inside them
 - AlltoAll completes only when the slowest rank has received everything
- On real workloads, what decides P99/P50 latency?

The implicit “MoE routing is uniform”

- Nobody has written this down but still baked in at three levels
- Instruments: nccl-tests / OSU issue a uniform AllToAll, by construction
- MoE systems work: synthetic streams
 - HeterMoE’s random tensors, Tutel’s Zipf
- Trace formats: a Chakra a2a node carries ONE message size
 - p2p keeps the skew
- Downstream, two forces pull in opposite directions
 - Systems want more traffic inside the scale-up domain
 - Rail-only, hierarchical a2a
 - ML wants expert specialization, which IS cross-rank agreement on experts

The assumption, the hope, and the reality



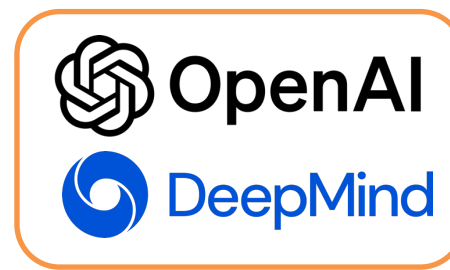
- What we measured instead: vertical stripes
 - Every source rank converges on the same few hot destinations
 - Not a defect: the training objective working exactly as intended
- A hot column is an incast, and expert-to-rank remapping cannot flatten it

Why this is hard to answer

- Three confounds travel together in the wild
 - Architecture (many open-weight models, all of different sizes)
 - Token stream (mock? Zipf or uniform? real text?)
 - Balancing config (auxiliary loss, bias terms)
- Our starting guesses, and how they broke
 - Hunch: the architecture's hidden-state geometry shapes the skew
 - Then swapping mock tokens for a Wikipedia corpus changed the story too
 - So: stop guessing, ablate real checkpoints with a factorial design

Choosing model architectures from the open zoo

- So many models, where should we actually look?
- What people do continued pretraining with
 - OLMo / Apertus (MHA, open-research favourite)
 - Gemma / GPT-OSS / Llama (overwhelmingly GQA)
- We also want to peek into frontier open-weight
 - DeepSeek / Kimi / GLM series (MLA-dominant)
- And to cover the emerging attention schemes
 - Nemotron-3 / Qwen-3.5 (SSM / gated linear attention)
- Our sequence-mixer array
 - Softmax attention: MHA, GQA, MLA
 - Linear attention: Mamba-2, Gated DeltaNet



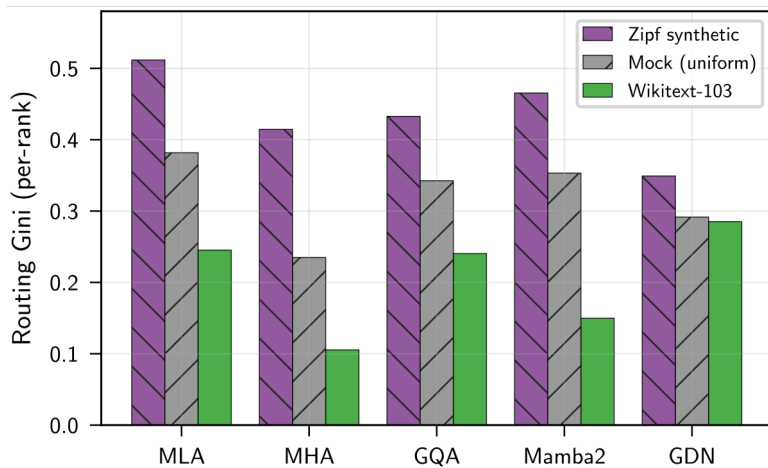
The observatory: 5 architectures × 6 data conditions

- Data: six conditions, from real text down to random IDs
 - Each one deletes a single ingredient of real data
- Scans: EP {4,8,16,32} in matched windows; GBS 64->2048, three seeds
- Megatron instrumentation: per-dispatch input splits on every rank
 - Every layer gives us the full S matrix per AlltoAll
 - Profiled on Helma: 4xH100 per node, NVLink intra-node, four NDR200 rails

Checkpoint	Mixer	Experts	Top-k
DeepSeek-MoE-16B	MHA (Multi Head Attention)	64	6
DeepSeek-V2-Lite	MLA (Multi-Head Latent Attention)	64	6
Qwen3-30B	GQA (Group-Query Attention)	128	8
Nemotron-30B	Mamba2 (Mamba-2 Hybrid)	128	8
Qwen3.5-35B	GDN (Gated DeltaNet Hybrid)	256	8

Mock tokens overestimate imbalance, everywhere

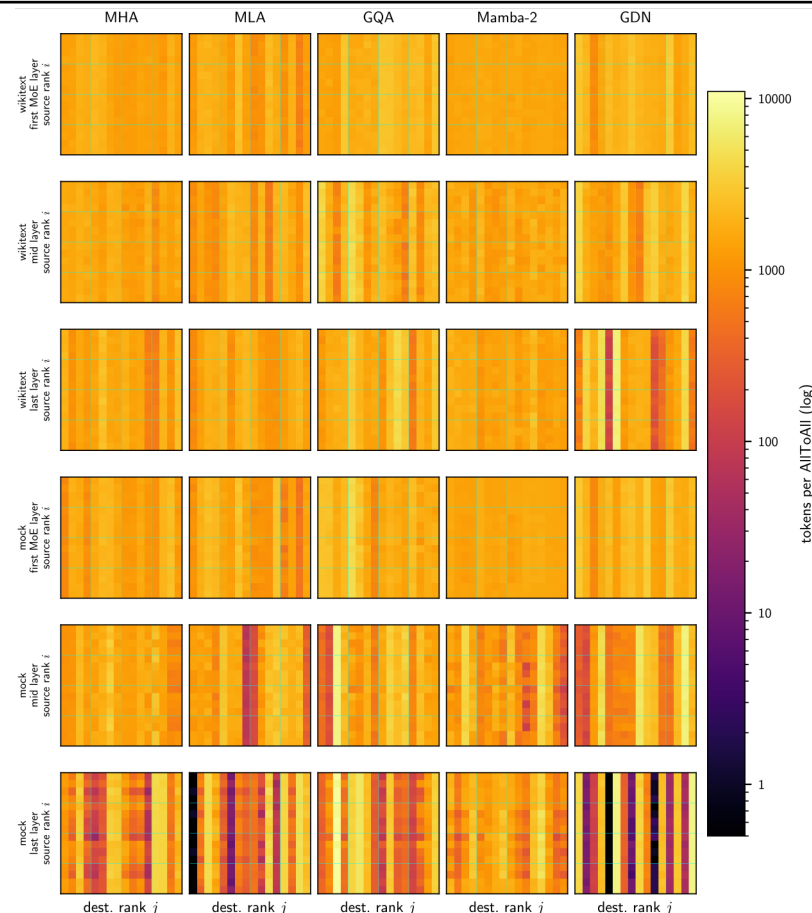
- Every architecture routes better on real text than on mock
 - Gini overestimated by up to 2.35x (Mamba-2)
 - Read as Dirichlet alpha, the same gap reaches 47x (0.16 -> 7.5)
- A scheduler tuned on mock over-provisions tail bandwidth
 - By roughly 2x for MHA-class models



Zipf makes things worse!

Where the imbalance lives in depth

- Mock data overestimates most severely in the deep layers
 - Fabricates idle ranks and stragglers
- Persistently concentrated: MLA, GDN
 - Concentrated from the very first MoE block
- Data-resilient: MHA, Mamba-2
 - Real text drives their routing near-uniform
 - Start near-uniform and concentrate only with depth
- GQA is the intermediate case
 - But BY DEPTH joins concentrated group



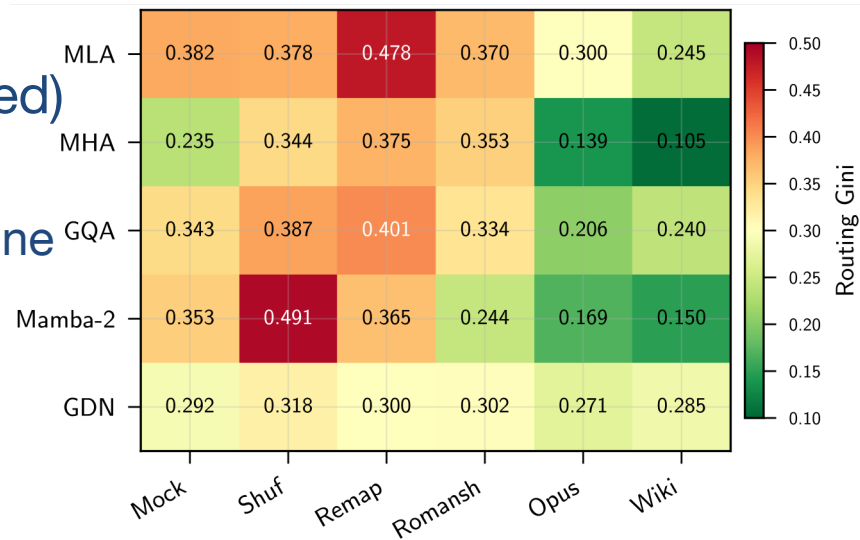
Dataset surgery: deleting one ingredient at a time

Condition	Looks like	What is deleted
wikitext	Whoever knows Romansh also knows German	Nothing; real, seen English
Opus	You are a CUDA expert debugging tcgen06 ...	“Seen it in pretraining”
Romansh	Tgi che sa rumantsch, sa er tudestg	Language familiarity
shuffled	German Whoever knows Romansh knows also	Word order (frequencies survive)
remapped	Est che sa tud er sa rum	Word meaning (shape survives)
mock	ants ch rum tud est g T	Everything (uniform-random IDs)
Zipf	the of and to is in a the is that was it for the	Everything except the frequency curve

- Romansh: ~60,000 speakers, Latin-derived, heavy German influence
 - Absent from standard pretraining corpora
- Whichever deletion breaks realistic routing names what the router actually reads

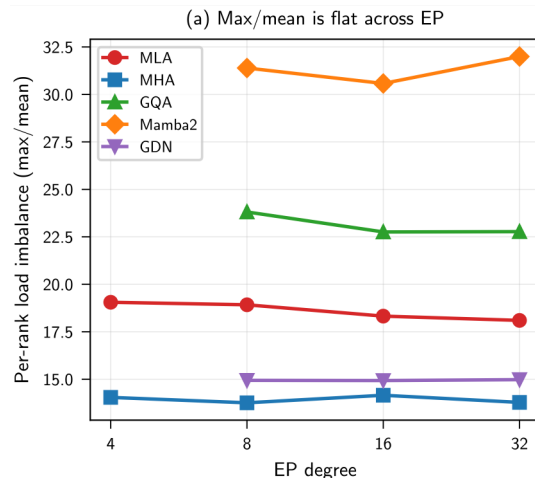
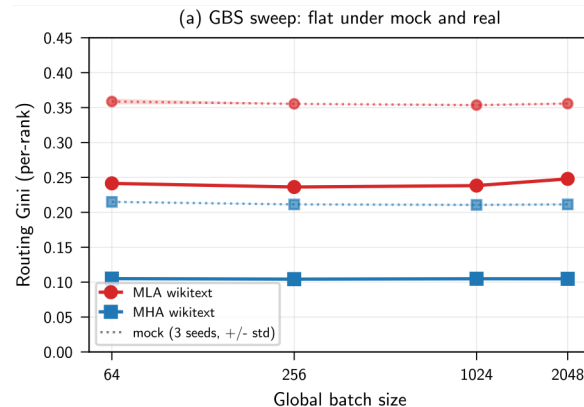
Dataset surgery: what the router actually reads

- Shuffled real tokens (sequence destroyed)
 - As bad as mock, or worse
- Remapped (embedding lookup destroyed)
 - Mock-tier or worse
 - At or above the model's own mock baseline for all five
- Routing is token-identity driven
 - It needs the learned embeddings and the sequence together
 - Matching one marginal (unigrams) moves you away from real routing, not toward it



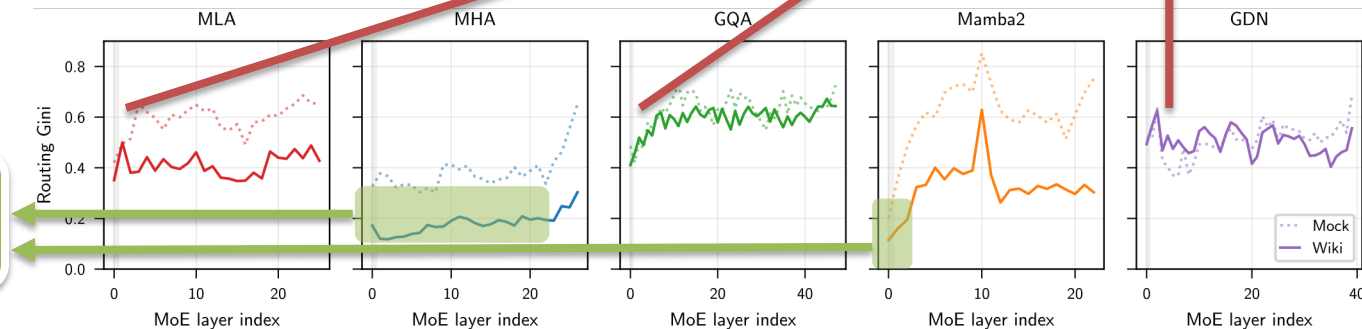
Two “fake” knobs: EP degree and Batch Size

- Does EP degree change the load imbalance?
 - Per-expert max/mean is flat ($\leq 5\%$) from EP=4 to EP=32
 - Flat straight across NVLink to InfiniBand bandwidth transition
 - EP changes where a hot expert lives
 - Not how hot it is
- Does a bigger, more varied batch help?
 - GBS 64 $\rightarrow$ 2048 (32x), three seeds
 - Imbalance flat within 3%, mock and real alike
 - Mock lies about the level, not the slope



Where can we force balance safely?

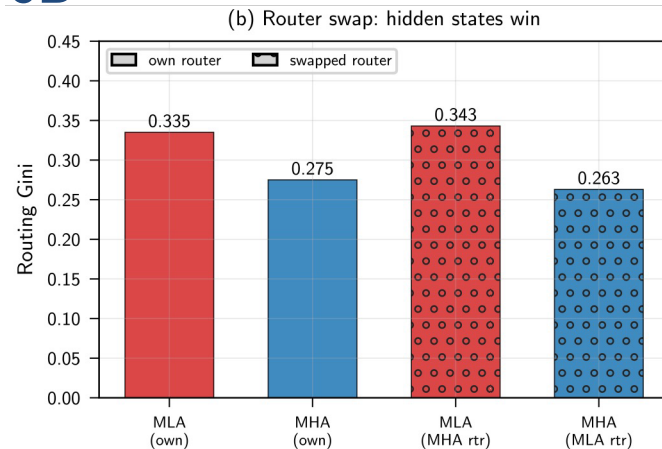
- Early-layer routing looks like a vocabulary partition
 - Something a hash function approximates well
 - cf. DeepSeek-V4's hash-routed early blocks
- Our data says for whom that substitution is safe
 - Resilient (MHA, Mamba-2)
 - ~~Concentrated (MLA, GDN, and GQA)~~



Early layer hash routing
is near-loss-free

Router weights, or attention block?

- DeepSeek-V2-Lite (MLA) and DeepSeek-MoE-16B (MHA) give us a rare control group
 - Identical MoE block: 64 routed + 2 shared
 - Top-6, same expert width, same aux loss
 - Attention mechanism differs
- Swapped the two routers (mlp.gate.weight only)
 - Both are 64x2048, so the gate transplants
 - Body/experts/embeddings all stay
- Gini follows the body, not the router weights
 - The router is a lens: swap lenses, same picture
 - The skew is written into the hidden-state geometry the attention hands the router



Takeaways

- Routing skew is decided by model x data, EP degree is not the lever
- Mock data is miscalibrated, and Zipf does not fix it
- Architecture class shows how much room you have to force balance
 - Uniform-load design math is valid for one class and misleading for the other
- Rebalance from the load you actually see, do not predict it
 - Placement is fitted on past traffic but paid on future traffic
 - Drift eats 17-60% of the gain (our traces)
 - Perfect knowledge gets max/mean to 1.00-1.07
 - UltraEP already measures 1.01-1.04
- Scale-up raises the baseline, and that helps
 - But at production E/P replication is mandatory, not optional

Looking forward: 10 Trillion Parameter Era

- MoE is getting sparser, not denser
 - Sparsity sets a model-intrinsic incast ceiling
- Kimi-K3 already activates just 16 of 896 experts
 - Worst-case concentration $E/k = 896/16 = 56:1$
 - The next generation pushes the active fraction lower still
- A 10T model at a similar active-parameter budget
 - ~2-3x the expert count drives that ratio toward ~128:1
 - And as E/P approaches 1, the rank holding the hot expert takes a synchronized
 - Fan-in from every other rank at once
- Sparser is not cheaper on the wire
 - More experts per token, each message smaller (latent, quantized)
 - Binding constraint shifts from bandwidth -> message rate + buffer occupancy

Thank you!



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